CloudWatch Agent – Advanced EC2 Monitoring
To Begin with the Lab

Summary of the Lab

In this lab, an Amazon EC2 instance was launched and configured to use the CloudWatch Agent for advanced monitoring. An IAM role with the required permissions was created and attached to the EC2 instance so that the Systems Manager (SSM) Agent and CloudWatch Agent could function correctly. The CloudWatch Agent was installed and configured directly from the Amazon CloudWatch console to collect additional system-level metrics such as memory and disk utilization. Finally, the configuration was verified by confirming that the **CWAgent** namespace appeared in Amazon CloudWatch and that advanced metrics were successfully collected.

Understanding CloudWatch Agent in Amazon CloudWatch

The **CloudWatch Agent** is an AWS software agent that runs on Amazon EC2 instances and on-premises servers to collect operating system metrics and log files that Amazon CloudWatch cannot collect by default. It exists because the default EC2 monitoring only provides infrastructure metrics such as CPU utilization, network traffic, and disk I/O, while memory usage, disk space, swap usage, and process information require an additional agent. This enables administrators to monitor the complete health of their servers from a single CloudWatch dashboard. The result is faster troubleshooting, better visibility, and proactive alerting for both cloud and on-premises workloads.

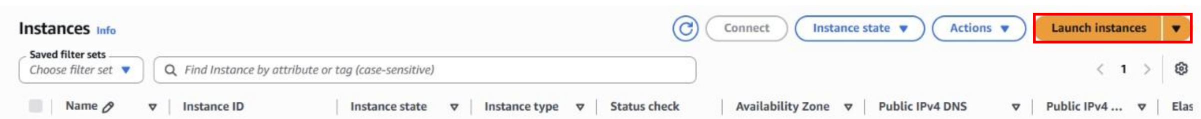
Example:

A production EC2 instance named **WebServer** hosts an e-commerce application where CPU utilization remains below 40%, but customers report slow response times. After installing the CloudWatch Agent, the administrator discovers that available memory has dropped below 10% and disk space on the application volume is nearly full. With these additional metrics, the root cause is identified quickly and resolved before the application becomes unavailable.

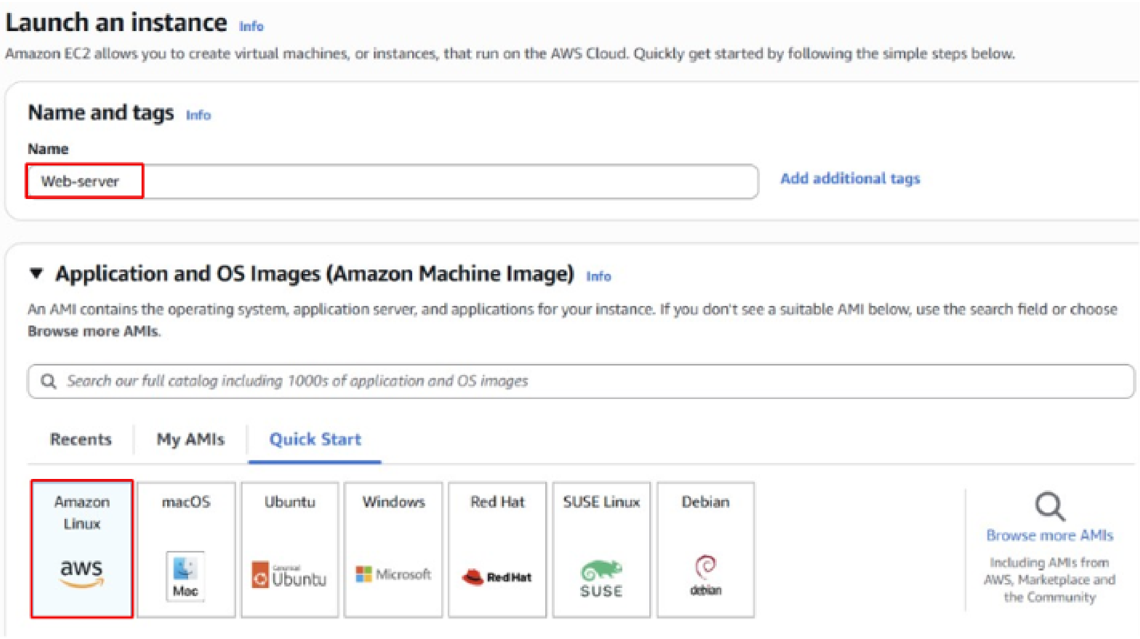
Lab Steps

Step 1: Launch an EC2 Instance

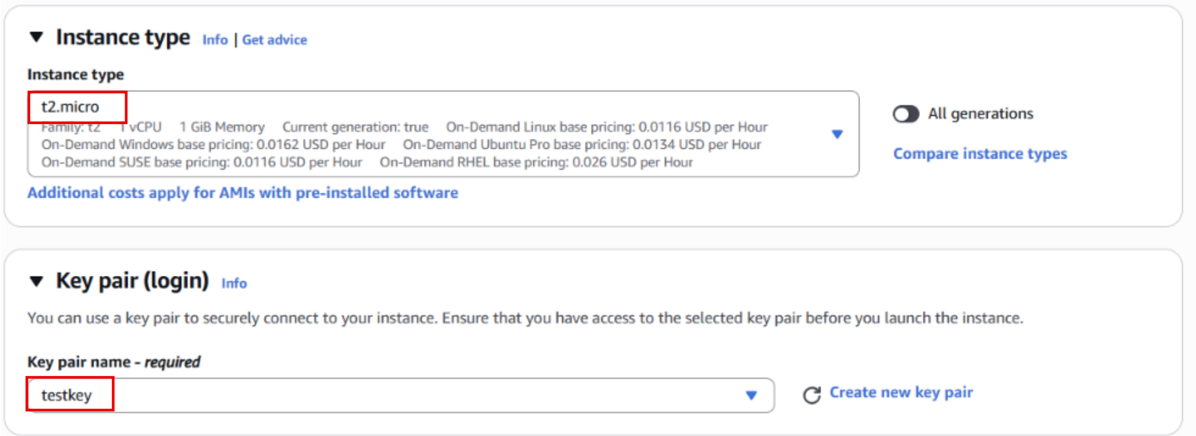
- Open the AWS Management Console.
- Go to **EC2**.
- Click **Launch Instance**.



- Enter the instance name (for example, **Web Server**).
- Select **Amazon Linux 2023 AMI**.

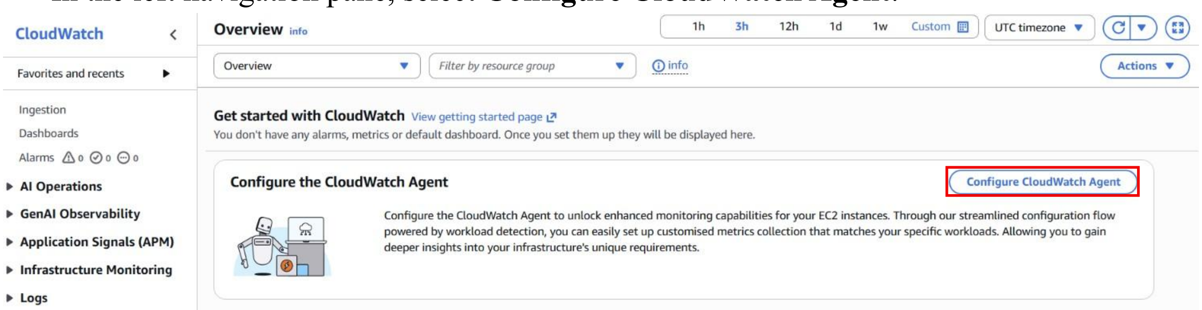


- Choose **t2.micro** (or **t3.micro**) as the instance type.
- Select an existing key pair.
- Keep the default settings.
- Click **Launch Instance**.



Step 2: Open the CloudWatch Console

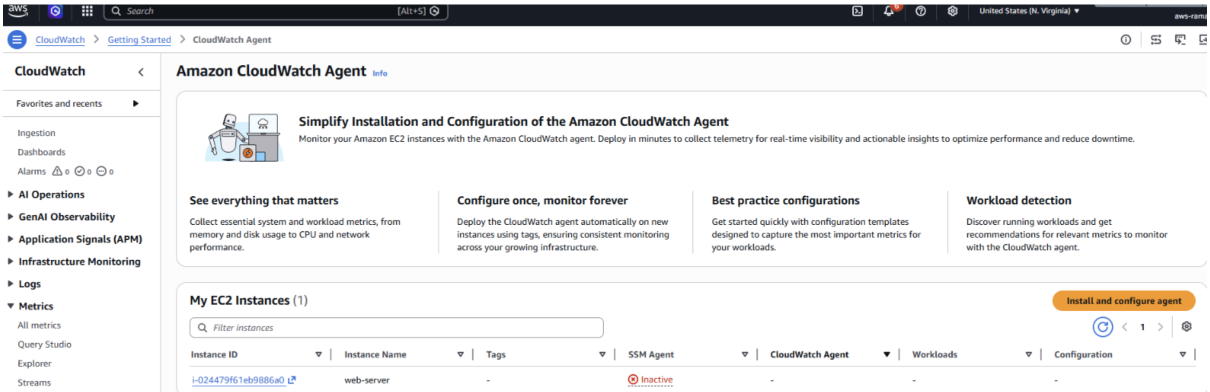
- Go to the **CloudWatch** service.
- In the left navigation pane, select **Configure CloudWatch Agent**.



- Wait until your EC2 instance appears in the list.

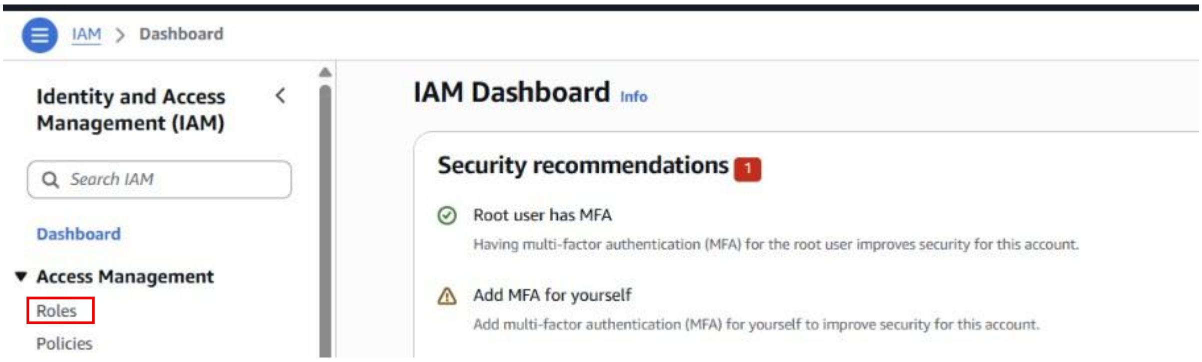
Step 3: Check the SSM Agent Status

- Locate your EC2 instance.
- Verify the **SSM Agent Status**.
- If the status is **Inactive**, an IAM role is required before CloudWatch Agent can be installed.

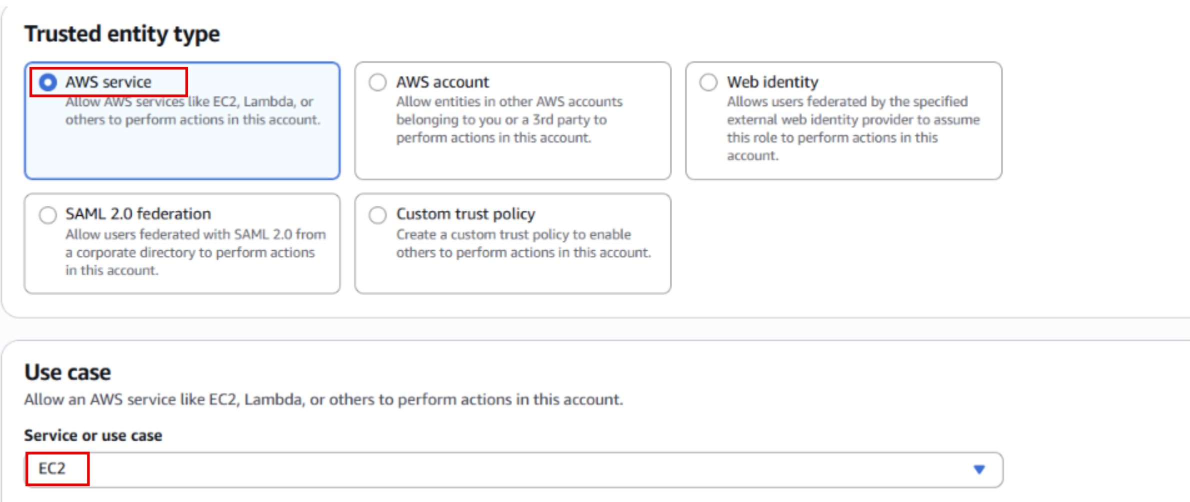


Step 4: Create an IAM Role

- Go to the **IAM** service.
- Select **Roles**.



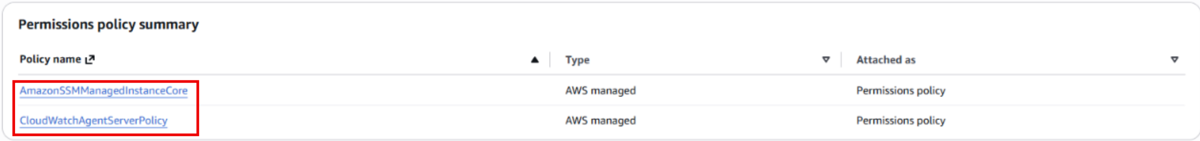
- Click **Create Role**.
- Choose **AWS Service**.
- Select **EC2** as the trusted service.



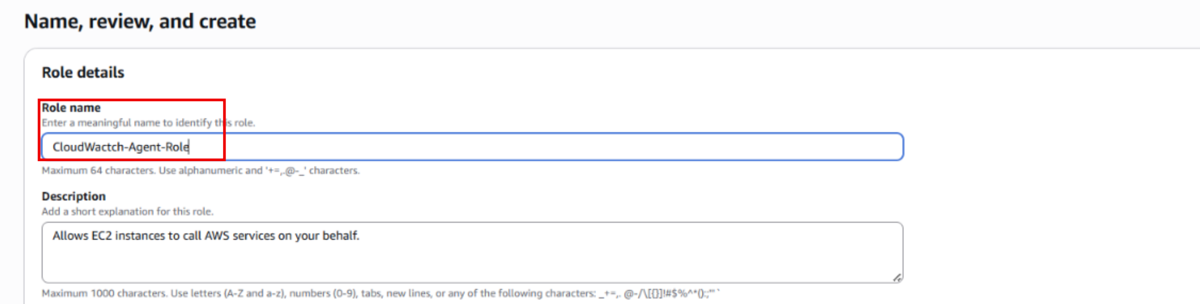
- Click **Next**.

Step 5: Attach Required IAM Policies

- Search for and attach **AmazonSSMManagedInstanceCore**.
- Search for and attach **CloudWatchAgentServerPolicy**.



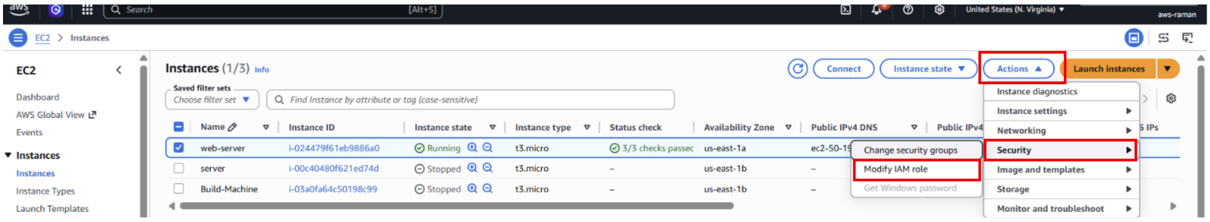
- Click **Next**.
- Enter the role name (for example, **CloudWatch-Agent-Role**).



- Click **Create Role**.

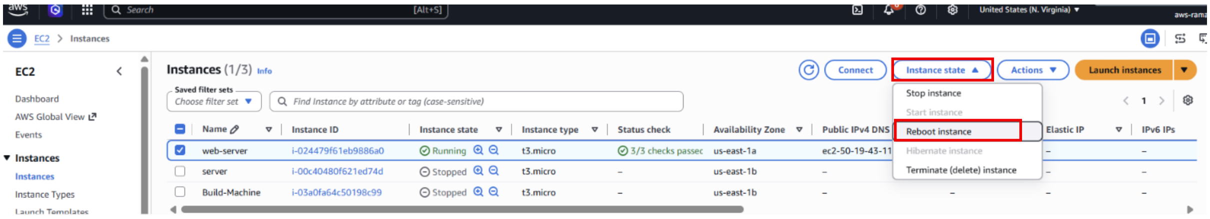
Step 6: Attach the IAM Role to the EC2 Instance

- Return to the **EC2** console.
- Select your EC2 instance.
- Choose **Actions → Security → Modify IAM Role**.
- Select **CloudWatch-Agent-Role**.
- Click **Update IAM Role**.

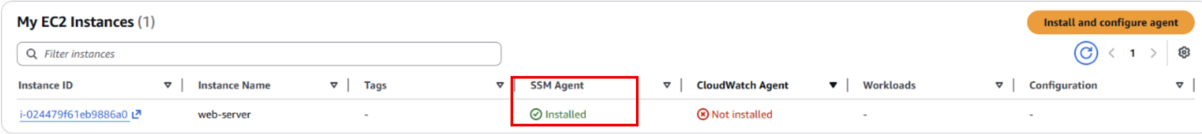


Step 7: Reboot the EC2 Instance

- Select the EC2 instance.
- Choose **Instance State → Reboot Instance**.

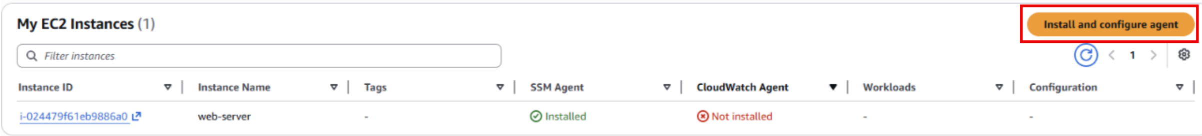


- Wait until the instance restarts.
- Return to the **CloudWatch** console.
- Confirm that the **SSM Agent Status** is now **Installed/Active**.

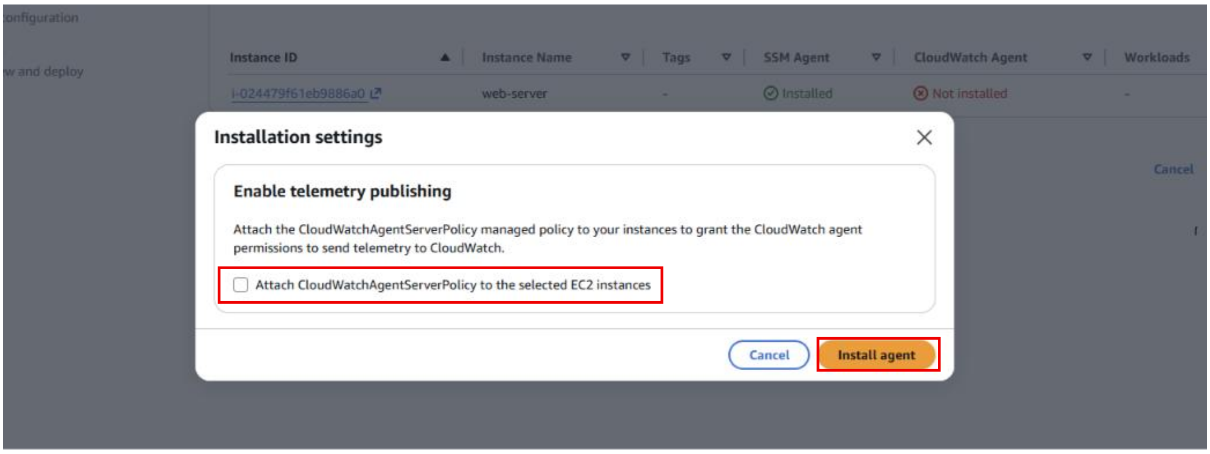


Step 8: Install the CloudWatch Agent

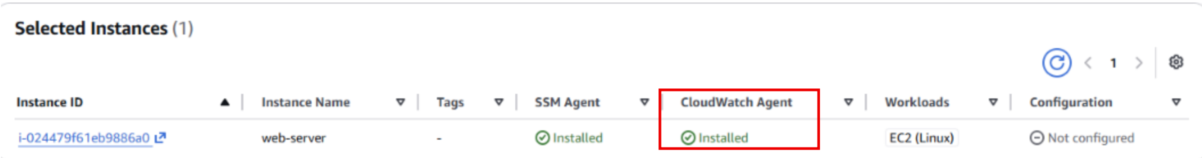
- In **CloudWatch**, click **Install and Configure Agent**.



- Select the EC2 instance.
- Click **Next**.
- Disable CloudWatchAgentServerPolicy
- Click **Install Agent**.

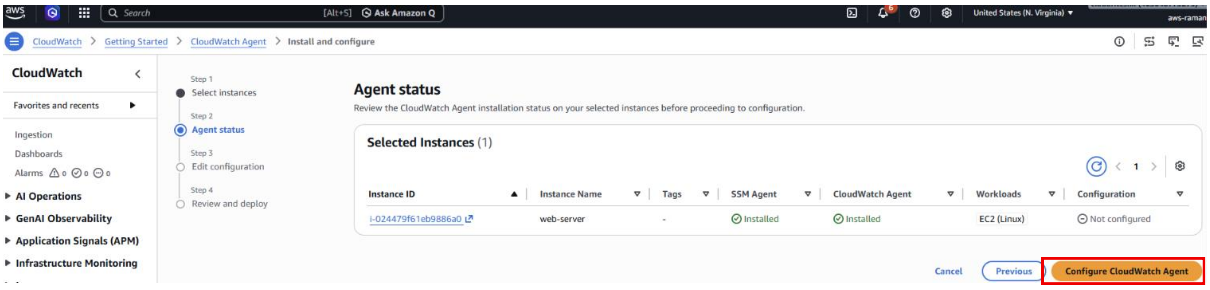


- Wait until the installation completes successfully.



Step 9: Configure the CloudWatch Agent

- Click **Configure CloudWatch Agent**.



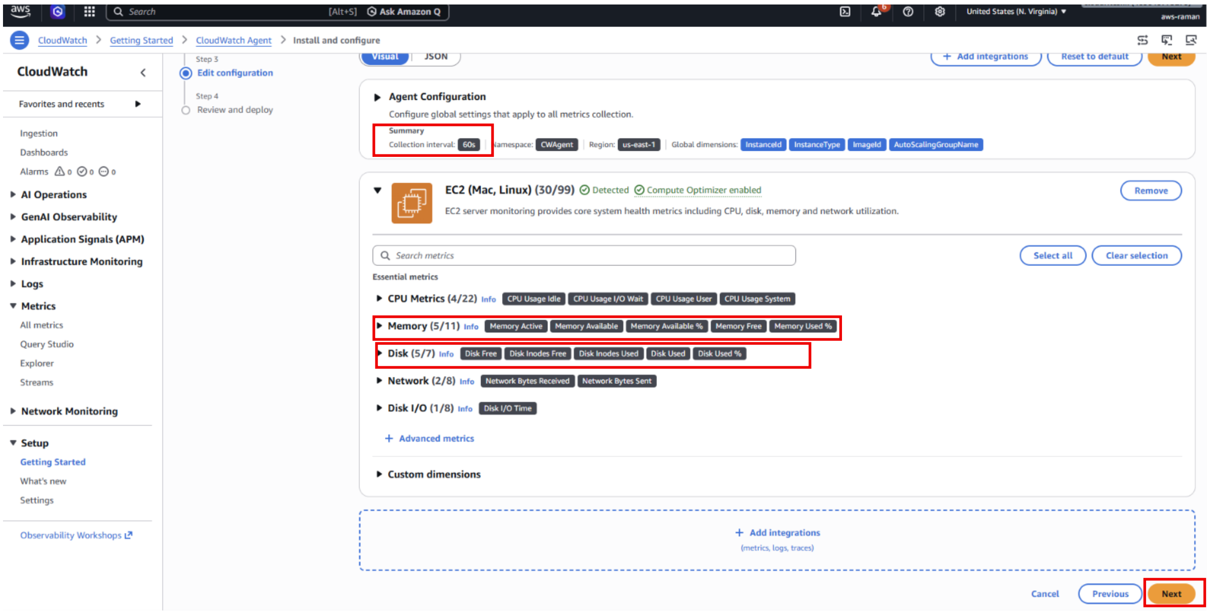
- Keep the default collection interval (**60 seconds**).
- Verify the metric namespace as **CWAgent**.
- Choose the metrics you want to collect, such as:

  1.Memory Available

  2.Memory Available Percentage

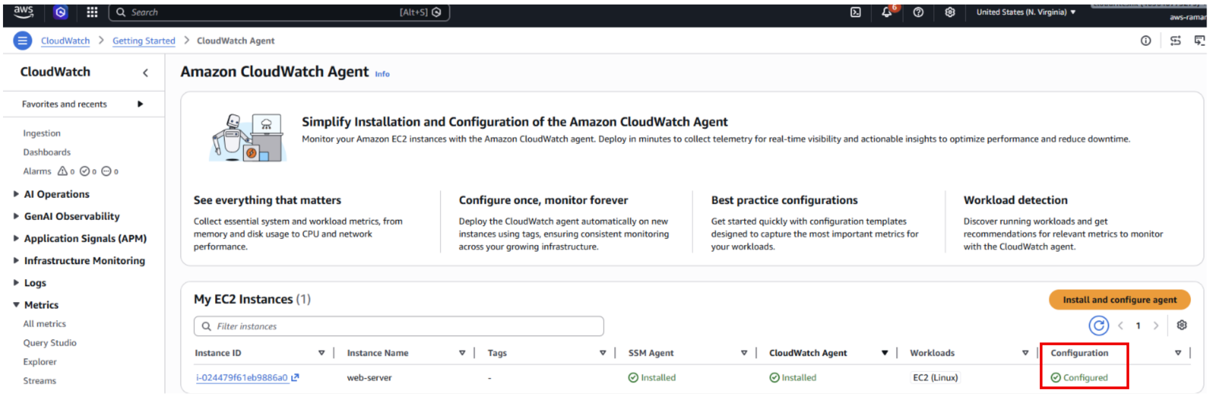
3.Memory Free
4.Memory Active
5.Disk Free
6.Disk Usage
7.Running Processes

- Click **Next**.
- Click **Deploy**.



Step 10: Verify the Configuration

- Wait until the status changes to **Installed** and **Configured**.
- Refresh the CloudWatch page if required.



Step 11: View Advanced Metrics

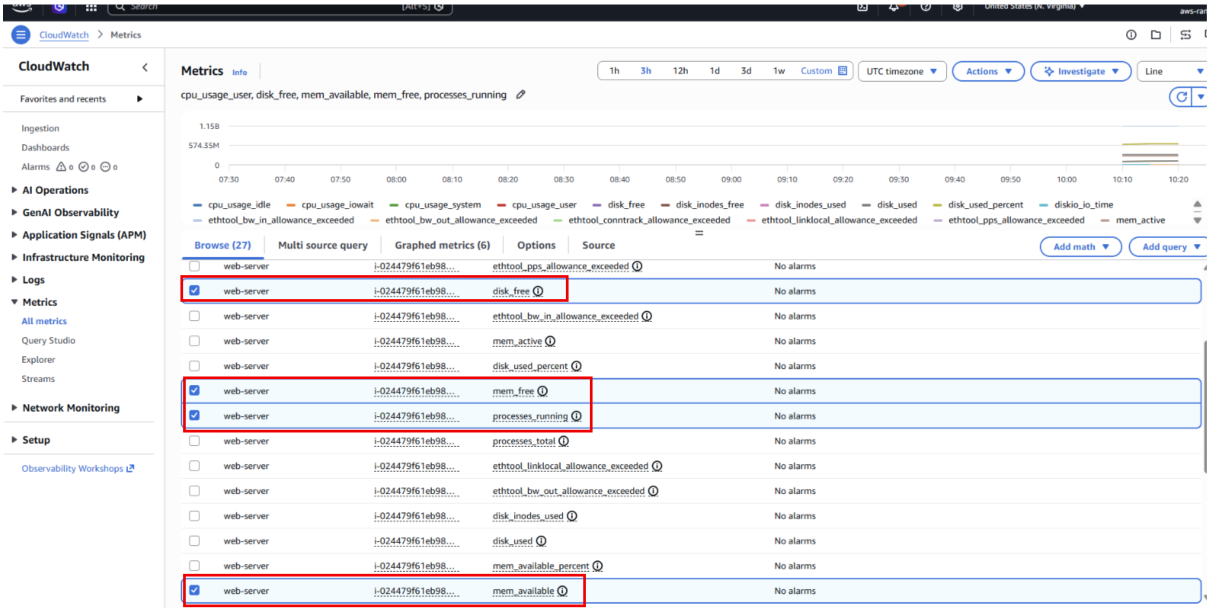
- Go to **CloudWatch → Metrics → All Metrics**.
- Open the **CWAgent** namespace.
- Select your EC2 instance.
- View advanced metrics such as:

1.Memory Usage
2.Memory Available
3.Memory Free
4.Disk Space

5.Running Processes

6.CPU Time

7.Other system-level metrics



Step 12: Monitor the Instance

- Wait **10–20 minutes** for metrics to populate.
- Run applications or generate workload on the EC2 instance.
- Observe the changes in the CloudWatch graphs.